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System and method for removal of PFAS and other emerging contaminant micro-constituents from landfill leachate and other impacted liquids

Abstract

A system and method for removing PFAS from residual waste streams generated during the processing of landfill liquids prior to disposal are provided. An example of a landfill liquid is leachate. Landfill leachate can contain dissolved or entrained environmentally harmful substances such as PFAS and other emerging contaminant micro-constituents that may enter the environment if not properly treated. The presently disclosed system and method can improve the removal of PFAS and other emerging contaminant micro-constituents in landfill leachate and/or landfill impacted liquids including stormwater and contaminated groundwater.

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Background/Summary

BACKGROUND 1. Related Applications (1) This application claims the benefit, and priority benefit, of U.S. Provisional Patent Application Ser. No. 63/093,421, filed Oct. 19, 2020, the disclosure and contents of which are incorporated by reference herein in their entirety.

2. Field of the Invention

(1) The presently disclosed subject matter relates generally to removal of undesired chemicals from industrial waste streams, and more specifically, to removal of per- and poly-fluoroalkyl substances (PFAS) and other emerging contaminant micro-constituents from leachate treatment waste streams.

3. Description of the Related Art

(2) In recent years, there has been an increased interest in studying the effects of per- and poly-fluoroalkyl substances on humans. These substances (often referred to as “PFAS”) are manufactured chemicals that can be found in various consumer and industrial products.

(3) PFAS are sometimes referred to as “forever chemicals” because the molecules are characterized by a chain of strong fluorine-carbon bonds which result in a highly stable and long persisting chemical. It is believed that long term human exposure to PFAS can increase the risk of cancer, increase cholesterol levels, affect the immune system, and interfere with hormone levels.

(4) It is known in the art to use various treatment efforts to attempt to remove these per- and poly-fluoroalkyl substances from industrial waste streams. However, many of these previous treatment efforts have proven to be cost prohibitive and inefficient, and have not provided satisfactory results.

(5) For example, some conventional treatment approaches use adsorptive media such as granular activated carbon, natural zeolites, or ion exchange media to encounter the micro-constituent of interest in order to remove it from the bulk liquid. In the case of landfill leachate, the bulk liquid contains many constituents that may interfere with the adsorption process through physical and/or chemical mechanisms. Additionally, the micro-constituent concentration can be hundreds of orders of magnitude lower than other constituents in the landfill leachate and/or impacted landfill liquids.

(6) In particular, organic molecules (grossly referred to and non-specifically measured as total organic carbon) can cause significant interference with the adsorptive media. This interference results in decreased removal efficiency, reduced micro-constituent assimilative capacity, accelerated exhaustion, increased media usage and higher operational costs in order to achieve the desired micro-constituent removals and effluent concentrations.

(7) Improvements in this field are therefore desired.

SUMMARY

(8) Various illustrative embodiments of a system and method for removing PFAS from residual waste streams generated during the processing of landfill liquids prior to disposal are disclosed herein.

(9) In certain illustrative embodiments, a method for removing per- and poly-fluoroalkyl

substances (PFAS) from an initial waste stream is provided. The initial waste stream comprising per- and poly-fluoroalkyl substances can be introduced into a primary treatment module, wherein the primary treatment module is configured to remove particulate and colloidal organic matter from the initial waste stream using one or more of chemical coagulation and flocculation followed by one or more of solids separation and straight filtration, to produce an effluent stream. The effluent stream can be introduced into a secondary treatment module comprising an adsorptive organic/clay based material. The secondary treatment module can produce a treated waste stream that is substantially devoid of per- and poly-fluoroalkyl substances, wherein no removal of organic carbon from the waste stream is performed between the primary treatment module and the secondary treatment module. The initial waste stream can be a leachate treatment stream. The treated waste stream can have an aggregate concentration for per- and poly-fluoroalkyl substances of 17 ng/L or less after continuous processing for about 83 days.

(10) In certain illustrative embodiments, a method of removing per- and poly-fluoroalkyl substances (PFAS) from an initial waste stream is provided. The initial waste stream comprising per- and poly-fluoroalkyl substances can be introduced into a primary treatment module, wherein the primary treatment module is configured to remove particulate and colloidal organic matter from the initial waste stream using one or more of chemical coagulation and flocculation followed by one or more of solids separation and straight filtration, to produce a primary effluent stream. The primary effluent stream can be introduced into a secondary treatment module comprising an adsorptive organic/clay based material to produce a secondary effluent stream. The secondary effluent stream can be introduced into a tertiary treatment module comprising a reverse osmosis filtration unit to produce a treated waste stream. The treated waste stream can be substantially devoid of per- and poly-fluoroalkyl substances, and no removal of organic carbon from the waste stream is performed between the primary treatment module and the secondary treatment module. The initial waste stream can be a leachate treatment stream. The treated waste stream can have an aggregate concentration for per- and poly-fluoroalkyl substances of 17 ng/L or less after continuous processing for about 83 days.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) A better understanding of the presently disclosed subject matter can be obtained when the following detailed description is considered in conjunction with the drawings and figures herein, wherein:
- (2) FIG. 1A is a flow diagram of an illustrative system and method with membrane pretreatment according to embodiments of the present disclosure.
- (3) FIG. 1B is a flow diagram of an illustrative system and method with solids pretreatment according to embodiments of the present disclosure.
- (4) FIG. 2A is a flow diagram of an illustrative system and method with dual TOC removal vessels and adsorption media vessels according to embodiments of the present disclosure.
- (5) FIG. 2B is a flow diagram of an illustrative system and method with single TOC removal and adsorption media vessels according to embodiments of the present disclosure.
- (6) FIG. 3 is a flow diagram of an illustrative system and method with membrane filtration followed by adsorption according to embodiments of the present disclosure.
- (7) FIG. 4 is a flow diagram of an illustrative system and method with membrane filtration followed by adsorption and reverse osmosis according to embodiments of the present disclosure.
- (8) While the presently disclosed subject matter will be described in connection with the preferred embodiment, it will be understood that it is not intended to limit the presently disclosed subject matter to that embodiment. On the contrary, it is intended to cover all alternatives, modifications,

and equivalents, as may be included within the spirit and the scope of the presently disclosed subject matter as defined by the appended claims.

DETAILED DESCRIPTION

(9) Various illustrative embodiments of a system and method for removing PFAS from residual waste streams generated during the processing of landfill liquids prior to disposal are described herein.

(10) An example of a landfill liquid is leachate. Landfill leachate can contain dissolved or entrained environmentally harmful substances such as PFAS and other emerging contaminant micro-constituents that may enter the environment if not properly treated.

(11) In certain illustrative embodiments, the presently disclosed system and method can improve the removal of PFAS and other emerging contaminant micro-constituents in landfill leachate and/or landfill impacted liquids (including stormwater and contaminated groundwater).

(12) In certain illustrative embodiments, the system and method can include a microconstituent (i.e., PFAS) treatment with upfront biological treatment at a landfill site. For example, an on-site leachate treatment plant (LTP) can utilize a membrane bioreactor (MBR) and reverse osmosis (RO) to facilitate microconstituent (i.e., PFAS) reduction. The treatment approach can be utilized on landfill sites that deploy some form of on-site leachate treatment achieving significant biodegradable organics and suspended solids (TSS) removal.

(13) In certain illustrative embodiments, the presently disclosed system and method can include a number of physical and chemical treatment unit processes operated in series that are intended to improve the removal of micro-constituents in landfill leachate and/or landfill impacted liquids (including stormwater and contaminated groundwater). For example, in certain illustrative embodiments, the presently disclosed system and method can provide gross particulate removal via solids coagulation and precipitation and/or bag or cartridge filtration, followed by organic carbon removal using granular activated carbon or other organic compound assimilative media as needed, and then followed by a micro-constituent selective media. In certain illustrative embodiments, the system and method can be applied to untreated liquids and/or liquids that have been previously processed through biological treatment where additional removal of suspended particulates and/or organic constituents that are grossly typically defined as, and/or measured as, total organic carbon is required to improve overall micro-constituent removal.

(14) Additionally, in certain illustrative embodiments, membrane filtration can serve as a pre-filtration process ahead of the adsorptive media to remove solids and colloidal organic molecules that could blind media adsorption sites thereby reducing the PFAS removal capacity and increasing media usage/replacement rates and/or accumulate in the voids spaces, slowing the flow or causing channeling with the media bed, thus leading to an increase frequency of media bed backwashing.

(15) Illustrative embodiments of a system **10** for removing per- and poly-fluoroalkyl substances (PFAS) and other microconstituents from a waste stream are shown in FIGS. **1A** and **1B**. The waste stream can be, for example, a leachate stream that has been pre-treated.

(16) Referring now to FIGS. **1A** and **1B**, the system **10** can include a first treatment module (not shown) that is configured to perform a combination of chemical coagulation and flocculation to remove particulate and colloidal organic matter, followed by solids separation and/or direct filtration (media, bag, screen, membrane). The resulting material is delivered to a pretreatment effluent tank **25**. In FIG. **1A**, the pretreatment effluent tank **25** stores membrane pretreatment effluent, while in FIG. **1B**, the pretreatment effluent tank **25** stores solids pretreatment effluent. The system **10** can also include a second treatment module **35** that receives the materials from pretreatment effluent tank **25**, and that comprises one or more granular activated carbon (GAC) vessels configured to perform additional removal of total organic carbon (TOC). In embodiments where two vessels are utilized (see, e.g., FIG. **1A**), a lead/lag orientation can be implemented. The system **10** can also include a third treatment module **45** comprising one or more of: (i) an adsorptive media treatment (see FIG. **1A** and FIG. **1B**) such as specialty GAC or organic/clay

materials with or without physical/chemical surface modifications and configured to enhance micro-constituent constituent removal; and (ii) an ion exchange media resin treatment (not shown) configured to provide targeted micro-constituent removal. In embodiments where two vessels are utilized (see, e.g., FIG. 1A), a lead/lag orientation can be implemented. The embodiments of FIG. 1A and FIG. 1B each utilize a two vessel, lead/lag configuration, but are displayed differently for purposes of illustration herein. The product from the third treatment module **45** can be delivered to an RO influent tank (not shown) for further downstream treatment, or discharged for final disposal.

(17) In accordance with FIG. 1A and FIG. 1B, a method (corresponding to system **10**) for removing per- and poly-fluoroalkyl substances (PFAS) and other microconstituents from a waste stream is also provided. In certain illustrative embodiments, previously untreated liquids can be collected and transferred through a series of treatment steps. First, an initial treatment step is performed in a first treatment module (not shown) whereby particulate and colloidal organic matter is removed using a combination of chemical coagulation and flocculation followed by solids separation and/or straight filtration (media, bag, screen, membrane) and then delivered to a pretreatment effluent tank **25**. Second, the effluent stream from step one is pumped through a second treatment module **35** wherein one or more granular activated carbon (GAC) vessels facilitate additional removal of total organic carbon (TOC). Third, the GAC effluent is processed through a third treatment module **45** comprising one or more of: (i) an adsorptive media such as specialty GAC or organic/clay materials with or without physical/chemical surface modifications to enhance micro-constituent constituent removal; and (ii) an ion exchange media resin for targeted micro-constituent removal. In embodiments where two vessels are utilized, a lead/lag orientation can be implemented. Finally, the effluent from the third treatment module **45** is delivered to an RO influent tank (not shown) for further downstream treatment, or discharged for final disposal.

(18) Another illustrative embodiment of a system **20** for removing per- and poly-fluoroalkyl substances (PFAS) and other microconstituents from a waste stream is shown in FIG. 2A and FIG. 2B. The system **20** includes a PFAS and other microconstituent treatment for raw leachate that is low strength and not needing biological treatment. This treatment approach can address treatment needs for low strength raw leachate at sites that do not operate on-site LTPs.

(19) Referring now to FIG. 2A and FIG. 2B, the system **20** can include a raw leachate storage tank **65** for storing raw leachate from a landfill. The raw leachate can be fed to a series of TSS removal steps including a gross TSS removal vessel **60** that can be, e.g., bag filtration or coagulation and/or a fine TSS/colloidal removal vessel **55** that can comprise, e.g., macro-filtration. The product of these one or more TSS removal steps can be delivered to a second treatment module **35** comprising one or more granular activated carbon (GAC) vessels configured to perform additional removal of total organic carbon (TOC). In embodiments where two vessels are utilized (see, e.g., FIG. 2A), a lead/lag orientation can be implemented. The embodiments of FIG. 2A and FIG. 2B each utilize a two vessel, lead/lag configuration, but are displayed differently for purposes of illustration herein. The system **20** can also include a third treatment module **45** to receive the materials from second treatment module **35** and comprising one or more of: (i) an adsorptive media treatment such as specialty GAC or organic/clay materials with or without physical/chemical surface modifications and configured to enhance micro-constituent constituent removal; and (ii) an ion exchange media resin treatment configured to provide targeted micro-constituent removal. In embodiments where two vessels are utilized, a lead/lag orientation can be implemented. A product of the third treatment module **45** is a treated final effluent stream that is substantially devoid of PFAS and emerging contaminant microconstituents.

(20) In accordance with FIG. 2A and FIG. 2B, a method (corresponding to system **20**) for removing per- and poly-fluoroalkyl substances (PFAS) and other microconstituents from a waste stream is also provided. In certain illustrative embodiments, liquids can be collected and transferred through a series of treatment steps. First, an initial treatment step is performed in a first treatment module **25** whereby captured solids are removed using a combination of gross TSS removal and fine

TSS/colloidal removal. Second, the effluent stream from step one is pumped through a second treatment module **35** wherein one or more granular activated carbon (GAC) vessels for additional removal of total organic carbon (TOC). Third, the GAC effluent is processed through a third treatment module **45** comprising one or more of: (i) an adsorptive media such as specialty GAC or organic/clay materials with or without physical/chemical surface modifications to enhance micro-constituent removal; and (ii) an ion exchange media resin for targeted micro-constituent removal. In embodiments where two vessels are utilized (see, e.g., FIG. 2A), a lead/lag orientation can be implemented. Finally, the effluent from the third treatment module **45** is discharged for final disposal.

(21) Another illustrative embodiment of a system **30** for removing per- and poly-fluoroalkyl substances (PFAS) and other microconstituents from a waste stream is shown in FIG. 3. The system **30** can include a holding tank **80** for storing raw leachate from a landfill and/or pretreated leachate. The leachate from the holding tank **80** can be fed to a solids removal vessel **70** that can perform solids removal via membrane filtration (e.g., nano- or micro-filtration) on the leachate. The captured solids can be stored in a concentrate storage tank **75**, while the product stream can be delivered to a treatment module **45** comprising one or more of: (i) an adsorptive media treatment such as specialty GAC or organic/clay materials with or without physical/chemical surface modifications and configured to enhance micro-constituent removal (see FIG. 3); and (ii) an ion exchange media resin treatment (not shown) configured to provide targeted micro-constituent removal. In embodiments where two vessels are utilized (see, e.g., FIG. 3), a lead/lag orientation can be implemented. The product of the treatment module **45** is a treated effluent stream that is substantially devoid of PFAS and emerging contaminant microconstituents.

(22) In accordance with FIG. 3, a method (corresponding to system **30**) for removing per- and poly-fluoroalkyl substances (PFAS) and other microconstituents from a waste stream is also provided. In certain illustrative embodiments, liquids can be collected and transferred through a series of treatment steps. First, leachate from, e.g., a holding tank, can be fed to a solids removal vessel **70** that can perform solids removal via membrane filtration (e.g., nano- or micro-filtration) on the leachate. The captured solids can be stored in a concentrate storage tank **75**, while the product stream can be delivered to a treatment module **45** comprising one or more of: (i) an adsorptive media treatment such as specialty GAC or organic/clay materials with or without physical/chemical surface modifications and configured to enhance micro-constituent removal; and (ii) an ion exchange media resin treatment configured to provide targeted micro-constituent removal. In embodiments where two vessels are utilized, a lead/lag orientation can be implemented. The product of the treatment module **45** is a treated effluent stream that is substantially devoid of PFAS and emerging contaminant microconstituents.

(23) Importantly, in certain illustrative embodiments, the system **30** and corresponding method of FIG. 3 do not include removal of total organic carbon (TOC) in a TOC removal vessel prior to adsorption.

(24) Another illustrative embodiment of a system **40** for removing per- and poly-fluoroalkyl substances (PFAS) and other microconstituents from a waste stream is shown in FIG. 4. This approach also does not include removal of total organic carbon (TOC) in a TOC removal vessel prior to adsorption. As with FIG. 3, the approach of FIG. 4 allows for more efficient sequestering and encapsulation of the PFAS on the adsorbent media as compared to adsorption of the PFAS from the RO concentrate and then attempting to encapsulate the media or even solidification of the RO concentrate.

(25) In certain illustrative embodiments, system **40** allows for removal of PFAS and other micro-constituents between solids removal processes and facilitates adsorptive media targeted removal rather than universally concentrating all constituents into a single stream and then attempting PFAS removal and/or destruction within the concentrated stream. It is believed that other micro-constituents (in addition to PFAS) where a selective media could be used would also benefit from

this approach as compared to conventional strategies.

(26) In certain illustrative embodiments, system **40** allows for targeted removal of PFAS or other micro-constituents which can be effectively achieved as compared to attempting removal of the PFAS from the stream in the presence of higher concentrations of non-target constituents that reduce the effectiveness of PFAS selective processes. PFAS removal upstream of a reverse osmosis (RO) treatment process mitigates the accumulation of PFAS in the residual concentrate stream thereby facilitating disposal of the liquid in situations when elevated PFAS concentrations would prevent convenient disposal. PFAS removal upstream of a RO process also mitigates the accumulation of PFAS in the residual concentrate stream to concentrations that could trigger future RCRA hazardous waste classification.

(27) In accordance with FIG. **4**, a method (corresponding to system **40**) for removing per- and poly-fluoroalkyl substances (PFAS) and other microconstituents from a waste stream is also provided. In certain illustrative embodiments, liquids can be collected and transferred through a series of treatment steps. First, leachate from, e.g., a holding tank **80**, can be fed to a solids removal vessel **70** that can perform solids removal via membrane filtration (e.g., nano- or micro-filtration) on the leachate. The captured solids can be stored in a concentrate storage tank **75**, while the product stream can be delivered to a treatment module **45** comprising one or more of: (i) an adsorptive media treatment (see, e.g., FIG. **4**) such as specialty GAC or organic/clay materials with or without physical/chemical surface modifications and configured to enhance micro-constituent removal; and (ii) an ion exchange media resin treatment (not shown) configured to provide targeted micro-constituent removal. In embodiments where two vessels are utilized, a lead/lag orientation can be implemented. The product of the treatment module **45** is sent to a reverse osmosis filtration module **85** for further treatment. The product of the reverse osmosis filtration module **85** is a treated effluent stream that is substantially devoid of PFAS and emerging contaminant microconstituents.

(28) FIGS. **1-4** herein illustrate exemplary systems and methods with a plurality of sequential, non-sequential, or sequence independent “steps” using the equipment shown or described herein. It should be noted that the system and method of FIGS. **1-4** are exemplary and may be performed in different orders and/or sequences as dictated or permitted by the equipment described herein, and any alternative embodiments thereof. Numerous arrangements of the various “steps” and equipment can be utilized. In addition, not all “steps” or equipment described herein need be utilized in all embodiments. However, it should be noted that certain particular arrangements of equipment and/or “steps” for the system and method as described herein are materially distinguishable from and provide distinct advantages over previously known technologies.

(29) Experimental Testing

(30) Given concerns with typical high strength landfill leachate matrices and their impact on media-based technologies that serve as polishing steps to treat microconstituents such as PFAS, bench testing was initially conducted on raw leachate from an on-site leachate treatment plant and reverse osmosis concentrate. In order to validate the performance impacts on untreated landfill leachate, further bench testing was conducted to assess the effectiveness of carbon media in pretreating leachate for bulk organic carbon removal and achieving gross PFAS reduction, in accordance with the presently disclosed systems and methods. The bench testing demonstrated promising results for PFAS reduction using adsorption media.

(31) Three different media types were evaluated in a bench-scale adsorbent isotherm test. The media evaluated included: (i) granular activated carbon; (ii) an exchange media (Evoqua PSR2+ and DEXSORB®+); and (iii) an organic clay-based product (CETCO FluoroSorb®)

(32) At the completion of the reaction time (24 hours), total PFAS and PFOA concentrations of the bulk solution from each reactor was measured and equilibrium adsorption modeling for each media was calculated.

(33) Modeling results are shown in Table 1 below:

(34) TABLE-US-00001 TABLE 1 Raw leachate equilibrium adsorption modeling results PFOA PFAS Q_{max} C_{sub.10} %-Rem Q_{max} C_{sub.10} %-Rem Adsorbent K_{sub.f} 1/n (µg/g) (µg/L) at C_{sub.10} K_{sub.f} 1/n (µg/g) (µg/L) at C_{sub.10} Calgon F400 2.2 × 10^{sup.-4} 0.021 0.22 0.73 75% 6.1 × 10^{sup.-4} 0.243 1.3 11.6 49% CETCO 8.3 × 10^{sup.-4} 0.380 1.2 0.12 96% 6.3 × 10^{sup.-4} 0.701 5.6 5.46 76% FluoroSorb Evoqua 7.9 × 10^{sup.-4} 0.268 1.0 <0.69 >76% 2.6 × 10^{sup.-3} 0.279 6.2 1.23 95% PSR2+ CRC Care 4.0 × 10^{sup.-4} 0.649 0.8 0.57 80% 2.2 × 10^{sup.-6} 2.48 4.5 12.8 43% MatCare Notes: Initial concentration of PFOA = 2.9 µg/L; initial concentration off all measured PFAS = 22.5 µg/L Q_{max} = calculated maximum adsorption capacity using Freundlich equilibrium model parameters C_{sub.10} is the equilibrium effluent concentration remaining in the leachate at 10 g/L adsorbent dose (maximum dose tested)

(35) These results show that the CETCO FluoroSorb® achieved the highest PFOA specific removal rate (Q_{max}=1.2 ug/g) and demonstrated similar total PFAS removal when compared to Evoqua PSR2+ which achieved 6.2 ug/g whereas the FluoroSorb® achieved a 5.6 ug/g. This suggested that the media was least impacted by other constituents in the leachate and thus indicating that under some conditions, organic carbon removal may not be needed.

(36) In conjunction with the previous testing, removal/destruction of the PFAS from RO concentrate was also evaluated in bench-scale tests to evaluate the potential of electro-chemical oxidation process for this application. A series of batch studies were conducted in which different treatment conditions were applied to determine potential optimum operating conditions to achieve the highest PFAS removal performance. The results demonstrated that the technology could achieve levels of PFAS removal, but did not achieve the treatment goals of 56 ng/L for the aggregated total concentration of 4 target constituents after 40 hours of reaction time as demonstrated in the results. As such, this particular strategy is considered not to be a viable approach at the observed treatment times for large scale commercial implementation for RO concentrate associated with processing landfill leachate.

(37) Large-scale pilot testing of the presently disclosed system and method was conducted to confirm the observations of the bench-scale equilibrium testing. Biologically treated, membrane (ultra-filtration) filtered leachate was continuously processed through different media vessel configurations that included: Scenario 1—Vessels 1 and 2 contained GAC and Vessel 3 contained ion exchange media, and Scenario 2—Vessels 1 and 2 contained FluoroSorb®. For these evaluations, the treatment goal was 56 ng/L as the aggregate concentration of 6 selected PFAS constituents—PFHpA, PFOA, PFNA, PFDA, PFHxS, and PFOS.

(38) Under the Scenario 1 configuration, the treatment goal (<target concentration of 56 ng/L) was achieved for approximately 2 days. On day 3, the treatment goal for the final effluent (discharge from Vessel 3) was surpassed by approximately 1 ng/L for the aggregate concentration of the 6 target PFAS compounds.

(39) Under the Scenario 2 configuration, on day 83 the effluent from FluoroSorb® in Vessel 2 final effluent possessed an aggregate concentration for the 6 target PFAS compounds of 17 ng/L (target concentration was <56 ng/L). The effluent concentration (from the lead vessel) of the same 6 target constituents was 670 ng/L and could be allowed to go up higher to maximize constituent loading on the media in the lead column. The vessels were operated at typical operating conditions for the type of media used, with only a marginal change in pressure increase (<2 psi) across the entire system. This suggests that there was little or no buildup of material within the media beds, and thus we did not conduct any backwashing or other media bed maintenance during this evaluation period.

(40) Results are shown in Table 3 below:

(41) TABLE-US-00002 TABLE 3 Large pilot-scale evaluation of results of GAC followed by ion exchange resin removal of 6 target PFAS constituents Day of UFP Influent GAC#2 Effluent IXR Effluent Study (ng/L) (ng/L) (ng/L) 1 13,010 <1.77 <1.81 2 — 2304.5 34.9 3 — 2964.3 56.7 4 — 4362 155.4

(42) It was believed that non-specific/unidentified organic constituents collectively referred to and

measured as TOC would interfere with the specialty media and reduce the affinity and/or removal capacity for adsorption or ion exchange, and that the GAC would serve as a sacrificial media sponge to remove the TOC and improve the performance of the media. However, the tested media achieved significantly longer service life and higher removal capacity without the need to remove the TOC. Based on the significant performance improvement, the study was ended and a full-scale demonstration system was installed and continues to be operated.

(43) The presently disclosed system and method have a number of advantages over prior art technologies. For example, existing technologies using adsorption media-based systems offer direct application of their technology for PFAS reduction. However, the often-high strength concentrations and complexed matrix observed in landfill liquids (combination of organics and inorganics) results in inefficient performance of these technologies (especially adsorption-based media systems) thereby requiring some level of pretreatment to limit/minimize competing ions for media adsorption sites and allowing for more enhanced/efficient microconstituent (i.e. PFAS) removal. Further, existing technologies for micro-constituent(s) (i.e. PFAS or other emerging contaminants) have principally been demonstrated on water/wastewater matrices that are not as complex as landfill leachate and leachate impacted liquids. The chemical makeup and the relative concentrations of the different constituents in leachate have proven to be challenging for technologies developed to remove micro-constituents from less-complex matrices such as groundwater.

(44) The presently disclosed system and method improve micro-constituent removal efficiency, extend media service-life, and reduce operational costs. In certain illustrative embodiments, the system and method described herein utilize two adsorptive media such as activated carbon or other media that will target organic molecules (grossly defined as total organic carbon) followed by targeted media for the removal of micro-constituents such as PFAS or other emerging contaminants.

(45) In addition, the presently disclosed system and method improve the removal performance of micro-constituents in landfill leachate and other impacted liquids. Conventional treatment approaches use adsorptive media such as granular activate carbon, natural zeolites or ion exchange media to encounter the micro-constituent of interest in order to remove it from the bulk liquid. In the case of landfill leachate, the bulk liquid contains many constituents that may interfere with the adsorption process through physical and/or chemical mechanisms. Additionally, the micro-constituent concentration can be hundreds of orders of magnitude lower than other constituents in the landfill leachate and/or impacted landfill liquids. In particular, organic molecules (grossly referred to and non-specifically measured as total organic carbon) can cause significant inference to the adsorptive media. This interference results in lowered removal efficiency, reduced micro-constituent assimilative capacity, reduced time to exhaustion, increased media usage, and higher operational costs, in order to achieve desired micro-constituent removals and effluent concentrations.

(46) By comparison, the presently disclosed system and method seek to reduce the organic (particulate and dissolved molecules) component concentrations in the bulk liquid prior to processing the leachate and/or landfill impact liquids using a media that is selective for a specific micro-constituent. This way, the presently disclosed system and method will reduce the potential interferences of the selective media to improve performance of the media.

(47) Additionally, a process such as reverse osmosis (RO) concentrates all retained constituents (including PFAS constituents) into a small (relative to the primary bulk RO permeate flow) residual stream that will require subsequent disposal at a considerable expense. Alternatively, in theory the smaller more highly concentrated PFAS volume more readily facilitates the application of PFAS destruction technologies. However, given the retention of non-PFAS constituents in the RO residual, destructive technologies at the present time are not able to achieve low double digit PFAS concentrations in the final effluent at economically viable levels.

(48) In certain illustrative embodiments, the presently disclosed system and method is effective for removing and destroying specific emerging contaminant microconstituents that may be potentially concentrated in residual waste streams generated during the processing of the landfill liquids. In other words, these microconstituents can be removed and destroyed in addition to any PFAS that are removed and destroyed using the presently described technology.

(49) Certain existing leachate pretreatment systems allow raw leachate to flow into a storage tank and then get pumped into a treatment system that consists of, e.g., biological treatment followed by an ultra-filtration (UF) member system. Permeate from the UF membrane can be pumped through a reverse osmosis (RO) system to polish the permeate and meet discharge requirements. To the extent that PFAS would need to be removed from the RO concentrate, various technologies (e.g., electro-chemical oxidation and ion exchange specialty media) could achieve some removal, but do appear to be economically viable or sustainable for targeting low levels (<50 ng/L).

(50) In certain illustrative embodiments, the presently disclosed system and method is effective for preventing the PFAS from getting into the RO concentrate, which is beneficial because the RO concentrate will have other constituents that make it difficult for PFAS specific removal or destruction for targeting low levels. The presently disclosed system and method can target PFAS removal between UF (where there is a solids free liquid) and RO membranes.

(51) While the disclosed subject matter has been described in detail in connection with a number of embodiments, it is not limited to such disclosed embodiments. Rather, the disclosed subject matter can be modified to incorporate any number of variations, alterations, substitutions or equivalent arrangements not heretofore described, but which are commensurate with the scope of the disclosed subject matter.

(52) Additionally, while various embodiments of the disclosed subject matter have been described, it is to be understood that aspects of the disclosed subject matter may include only some of the described embodiments. Accordingly, the disclosed subject matter is not to be seen as limited by the foregoing description, but is only limited by the scope of the claims.

Claims

1. A method of removing per- and poly-fluoroalkyl substances (PFAS) from a leachate stream, the method comprising: introducing the leachate stream comprising per- and poly-fluoroalkyl substances into a primary treatment module, wherein the primary treatment module comprises a solids removal vessel that is configured to remove solids from the leachate stream using membrane filtration and produce an effluent stream; and passing the effluent stream directly from the solids removal vessel into a secondary treatment module comprising an adsorptive organic/clay based material and configured to adsorb microconstituents from the effluent stream, wherein there is no removal of total organic carbon (TOC) in a TOC removal vessel prior to adsorption occurring in the secondary treatment module, and wherein the secondary treatment module produces a treated leachate stream that is substantially devoid of per- and poly-fluoroalkyl substances.

2. The method of claim 1, wherein the membrane filtration is at least one of nano-filtration and micro-filtration.

3. The method of claim 1, wherein the secondary treatment module further comprises an ion exchange media resin treatment.

4. A method of removing per- and poly-fluoroalkyl substances (PFAS) from a leachate stream, the method comprising: introducing the leachate stream comprising per- and poly-fluoroalkyl substances into a primary treatment module consisting essentially of a solids removal vessel that is configured to remove solids from the leachate stream using membrane filtration and produce an effluent stream; and introducing the effluent stream directly into a secondary treatment module comprising an adsorptive organic/clay based material and configured to adsorb microconstituents from the effluent stream, wherein the secondary treatment module produces a treated leachate

stream that is substantially devoid of per- and poly-fluoroalkyl substances.

5. The method of claim 4, wherein the membrane filtration is at least one of nano-filtration and micro-filtration.

6. The method of claim 4, wherein the secondary treatment module further comprises an ion exchange media resin treatment.

7. A method of removing per- and poly-fluoroalkyl substances (PFAS) from a leachate stream, the method comprising: storing leachate comprising per- and poly-fluoroalkyl substances in a storage vessel; introducing leachate from the storage vessel directly into a primary treatment module using the leachate stream, wherein the primary treatment module consists essentially of a solids removal vessel that is configured to remove solids from the leachate stream using membrane filtration and produce an effluent stream; and passing the effluent stream directly from the primary treatment module into a secondary treatment module comprising an adsorptive organic/clay based material and configured to adsorb microconstituents from the effluent stream, wherein the secondary treatment module produces a treated leachate stream that is substantially devoid of per- and poly-fluoroalkyl substances.

8. The method of claim 7, wherein the membrane filtration is at least one of nano-filtration and micro-filtration.

9. The method of claim 7, wherein the secondary treatment module further comprises an ion exchange media resin treatment.

10. A method of removing per- and poly-fluoroalkyl substances (PFAS) from a leachate stream, the method comprising: storing leachate comprising per- and poly-fluoroalkyl substances in a storage vessel; introducing leachate from the storage vessel directly into a solids removal vessel using the leachate stream, wherein the solids removal vessel is configured to remove solids from the leachate stream using membrane filtration to produce an effluent stream; and passing the effluent stream directly from the solids removal vessel into an adsorption treatment vessel configured to adsorb microconstituents from the effluent stream using an adsorptive organic/clay based material, wherein the adsorption treatment vessel produces a treated leachate stream that is substantially devoid of per- and poly-fluoroalkyl substances.

11. The method of claim 10, wherein the membrane filtration is at least one of nano-filtration and micro-filtration.

12. The method of claim 10, further comprising passing the treated leachate stream through an ion exchange media resin treatment.
